

BP_GameTriggerArea

1. Component Overview

Purpose

BP_GameTriggerArea is the VR game trigger area Blueprint. It pulls game-rule configuration and trigger logic out of the GameManager: it detects the player entering/leaving the game area, drives the rule-selection UI, manages the countdown startup flow, and controls the VR character's UI interaction permission during gameplay.

Design

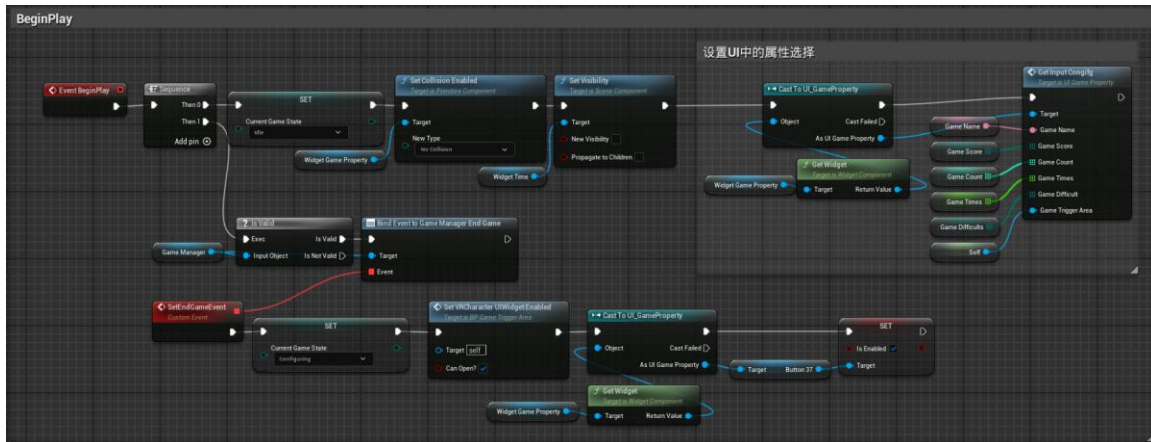
- Separation of responsibilities: rule configuration and trigger logic are pulled out of the GameManager, which only keeps the core game logic
- State-machine driven: the E_GameState enum (Idle / Configuring / Playing / End) manages the trigger area's lifecycle
- Data-driven UI: game properties (score mode, count, time, difficulty) are managed through arrays + indices, with left/right buttons switching options
- Countdown buffer: a countdown runs before the game starts to give the player preparation time; only after it finishes is the GameManager officially told to start
- UI interaction permission control: during gameplay the VR character's UI interaction component (BPC_03_VRUIWidget) is disabled so opening the UI mid-game cannot disturb the game

2. Initialization (BeginPlay)

Description

After the level loads, the trigger area initializes: sets the initial state, binds the game-end event, and fills the UI property data.

Blueprint screenshots



Key points

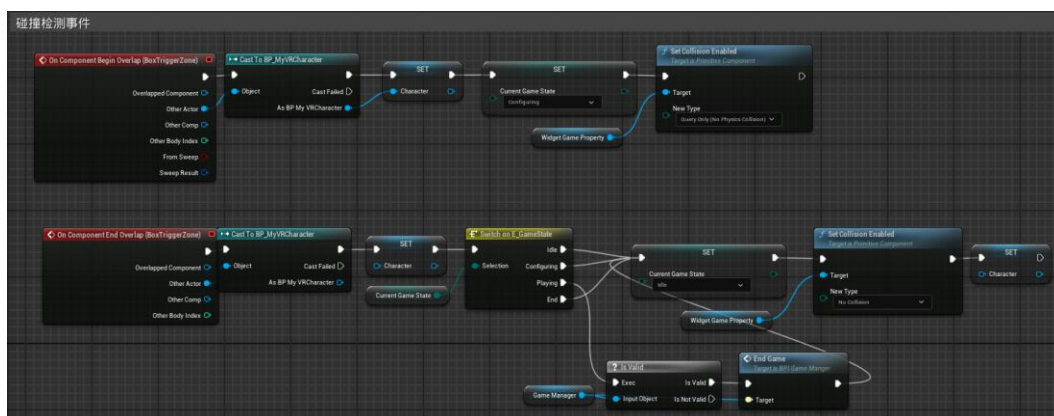
- Collision and visibility start disabled; they are enabled only when the player enters the area
- The End Game event is bound through a delegate, decoupling the GameManager from the TriggerArea
- Get Input Config passes the TriggerArea's own configuration data to the UI component
- After the game ends all UI interaction permissions are restored so the player can reconfigure and start the next round

3. Collision Events

Description

Detects the VR player entering/leaving the trigger area and runs different enter/leave logic depending on the current game state.

Blueprint screenshots



- Begin Overlap (player enters the area)
- End Overlap (player leaves the area)

Key points

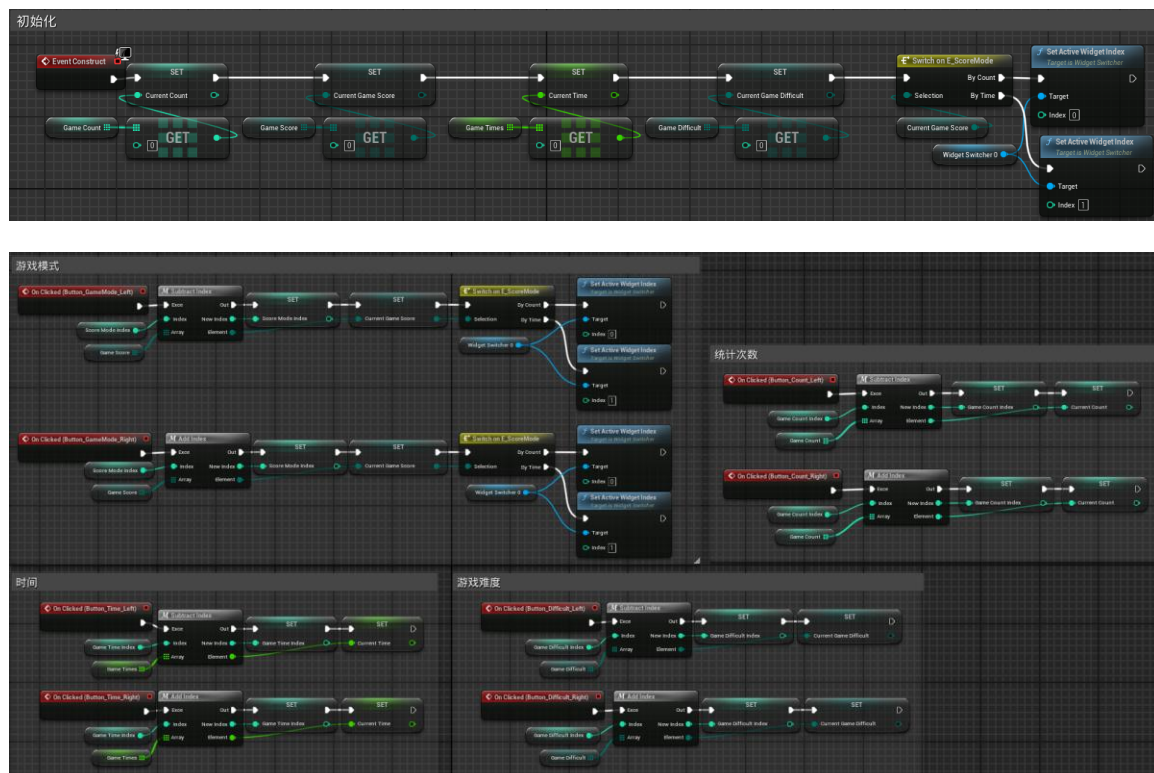
- Entering the area unconditionally sets the state to Configuring, because entering is exactly for configuring and starting the game
- Leaving in the Playing state calls End Game, because the game is running and the GameManager must be told to end it
- Leaving in the Idle / Configuring state does not call End Game, because the game has not started yet
- Leaving in the End state does not call End Game, because it has already been called
- The Character reference is cleared on leave to avoid dangling references

4. Game Property UI (WBP_UI_GameProperty)

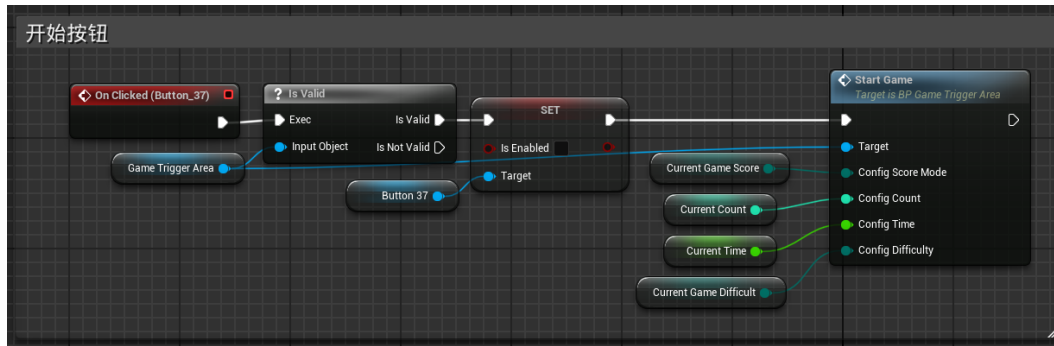
Description

Provides four groups of left/right toggle buttons for the player to pick the game-rule parameters: score mode, game count, countdown length, and difficulty. Once chosen, the Start button passes the configuration to BP_GameTriggerArea.

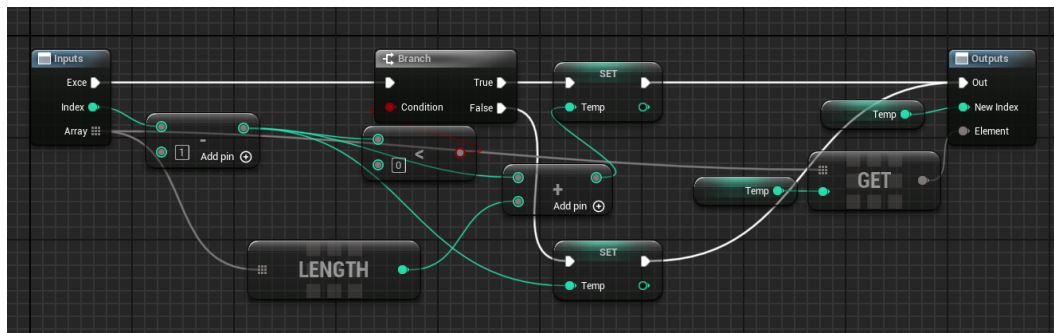
Blueprint screenshot - UI layout and button logic



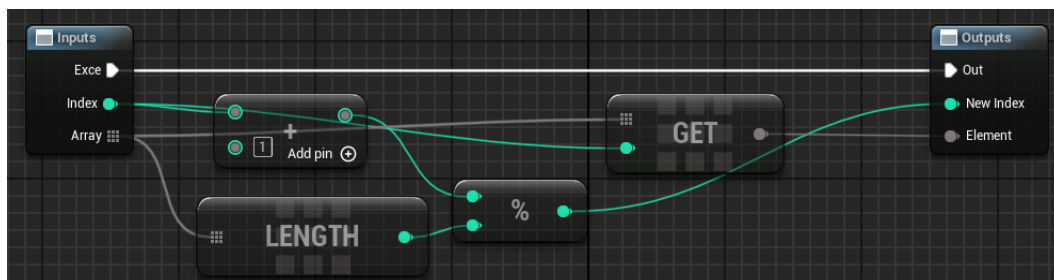
Blueprint screenshot - the Start button



Blueprint screenshot - left button index handling (decrement)



Blueprint screenshot - right button index handling (increment)



Four configuration groups

Option	Array Variable	Index Variable	Current Value Variable
Score mode	Game Score	Score Mode Index	Current Game Score
Game count	Game Count	Game Count Index	Current Count
Countdown length	Game Time	Game Time Index	Current Time
Game difficulty	Game Difficult	Game Difficult Index	Current Game Difficult

Key points

- The score mode drives a Widget Switcher; different modes show different sub-configuration panels

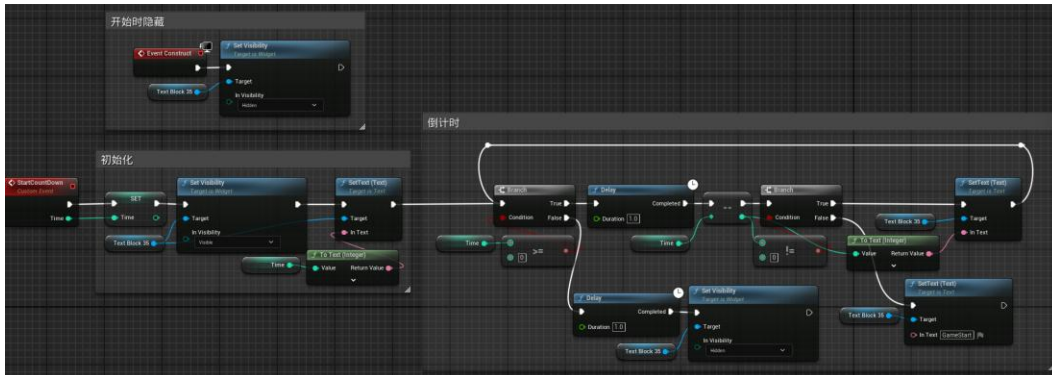
- Indices wrap around (not clamped), so the player can cycle through all options endlessly
- The Start button disables itself immediately after being clicked to prevent repeated StartGame calls
- Configuration is passed as function parameters rather than shared variables, keeping the data consistent

5. Countdown UI (WBP_UI_Countdown)

Description

Shows the countdown numbers (e.g. 3, 2, 1) before the game starts; when it ends, shows "GameStart!" to tell the player the game has officially begun. The countdown gives the player preparation time.

Blueprint screenshots



Key points

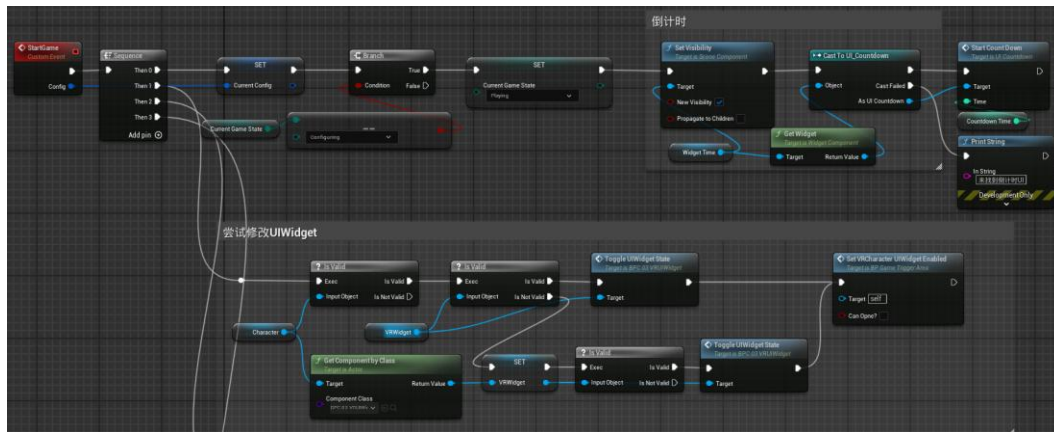
- When the TriggerArea sends the Start Game command to the GameManager in StartGame, the Delay parameter adds 1 extra second so the "GameStart!" text has enough time to display

6. StartGame Function

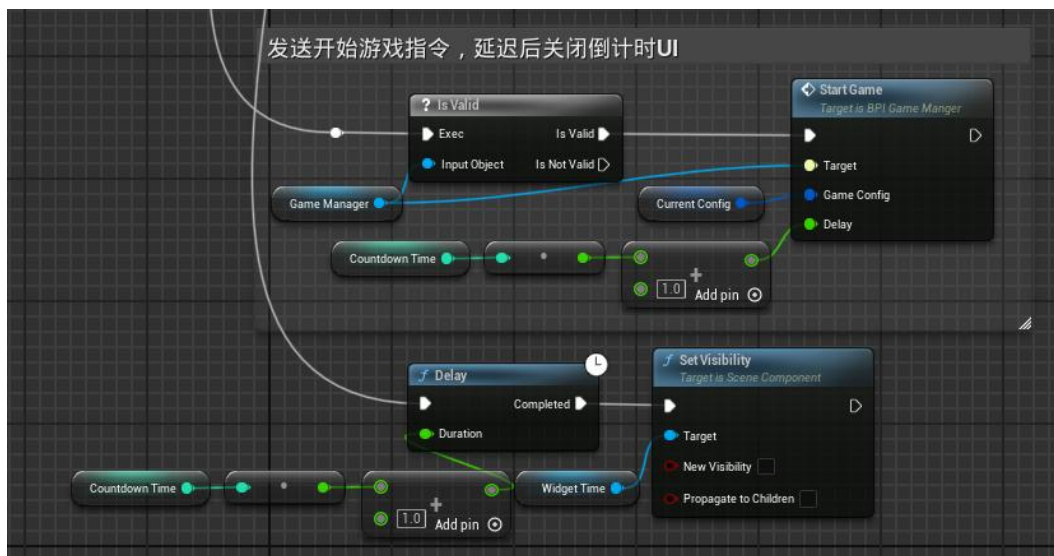
Description

Receives the game configuration from the UI, launches the countdown UI, disables the VR character's UI interaction, and notifies the GameManager to officially start the game once the countdown finishes.

Blueprint screenshot - StartGame main logic



Blueprint screenshot - sending the start command and closing the countdown UI after a delay



Key points

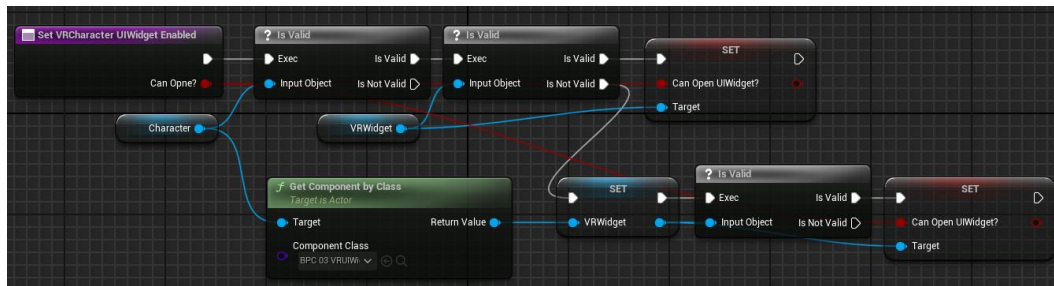
- The state only transitions to Playing from Configuring, preventing invalid transitions from other states
- Start Game is called through the BPI_GameManager interface; the Delay parameter tells the GameManager to wait until the countdown finishes before starting
- The VR character's UI interaction is disabled in parallel so rays from a UI opened mid-game cannot disturb the experience
- Both delays (Start Game and hiding the UI) use the same CountdownTime + 1.0 to stay in sync

7. Set VRCharacter UIWidget Enabled Function

Description

Controls the Can Open? boolean on the VR character's BPC_03_VRUIWidget component to disable/enable UI interaction during gameplay. The component reference is fetched lazily.

Blueprint screenshots



Key points

- Lazy loading: use the cache first and search for the component only when the cache is invalid, avoiding Get Component by Class every frame
- Double Is Valid checks: first the Character, then the VRWidget, preventing null-reference crashes
- The VRWidget variable is cached in the TriggerArea for direct use by later calls
- At game start Can Open? = false (disabled); at game end Can Open? = true (enabled)